5. Program 3: Library Management System

5.1 Introduction and Background

A Library Management System helps manage book records, availability, and borrowing activities. This project develops a console-based Python application that supports adding, viewing, issuing, returning, and searching books using in-memory data structures and a boxed terminal interface.

5.2 Objectives

- Develop a menu-driven library management system.
- Add and manage book records with unique IDs.
- Support book issue and return operations.
- Search books by ID, title, or author.
- Display book records and availability.
- Prevent invalid issue and return operations.

5.3 Problem Statement / Driving Question

How can a console-based application efficiently manage library records while preventing invalid book issue and return operations?

5.4 Methodology

- **Program Design:** Created a boxed terminal interface with reusable helper functions and an ASCII-art banner.
- **Book Record Management:** Stored book details in a `BOOKS` dictionary, implemented duplicate ID validation, and displayed records in a formatted table.
- **Issue, Return & Search Logic:** Developed functions to issue, return, and search books with proper validation and tested all menu operations.

5.5 Expected Outcomes / Deliverables

- A functional library management system.
- Accurate book issue and return tracking.
- Search functionality for book records.
- A clear, tabular catalogue display.
- Scope for future enhancements such as due dates and persistent storage.